



FABRIC MANUFACTURING II

Part: B

Course Code: FE 351-0723

Basic Stitches & Overlap, Underlap Variations

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LAPPING DIAGRAM

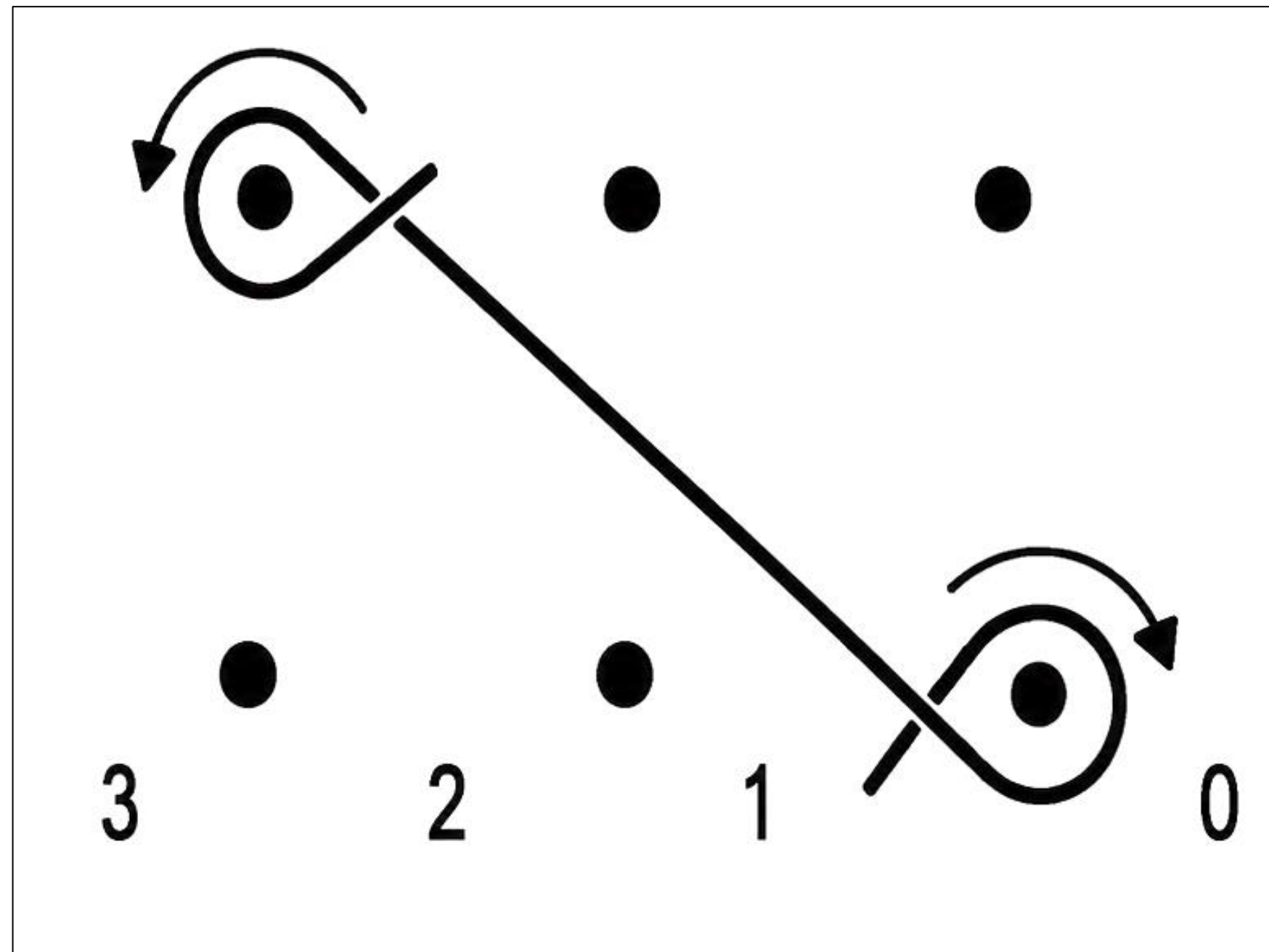
- Lapping diagram are drawn around *horizontal rows of points that represent needles in plain view.*
- Usually *assuming the pattern mechanism to be on the right.*
- As the guides position themselves in the *spaces between the needles* the positions between vertical columns of points can be given *chain link numbers commencing with the ‘O’ position which is to the right of the right-hand column of points.*
- Provided the direction and extent of the overlaps are correctly indicated in the lapping diagram and chain notation.

LAPPING DIAGRAM

When plotting a lapping diagram, a few basic rules must be observed:

- When the fabric is composed of *more than one guide bar, the lapping movement of each bar must be represented separately by one thread.*
- The lapping diagram of all bars must be plotted from the same course, so that the relative lateral position of all is kept.
- *The numbers allocated for the spaces between the needle must always correspond to the position of the pattern mechanism.*
- When mounting a chain on the pattern drum, special care must be taken to ensure the position of the chain corresponds to the movement of the knitting elements.

GUIDE BAR LAPPING DIAGRAM



CHAIN NOTATION

A Chain Notation is a list in correct sequence of chain link numbers spaced in to knitting cycles for each guide bar necessary to produce a particular structure.

O	U	O	U
1-0	/	2-3	/

O = Overlap and U = Underlap

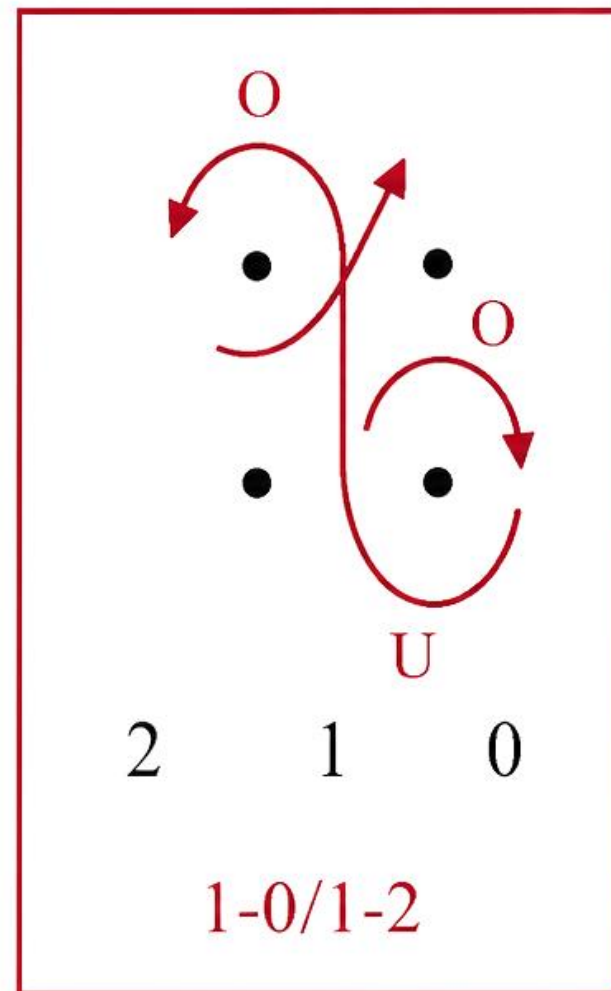
Overlap movements: 1 – 0, 2 – 3

Underlap movements: 0/2, 3/1

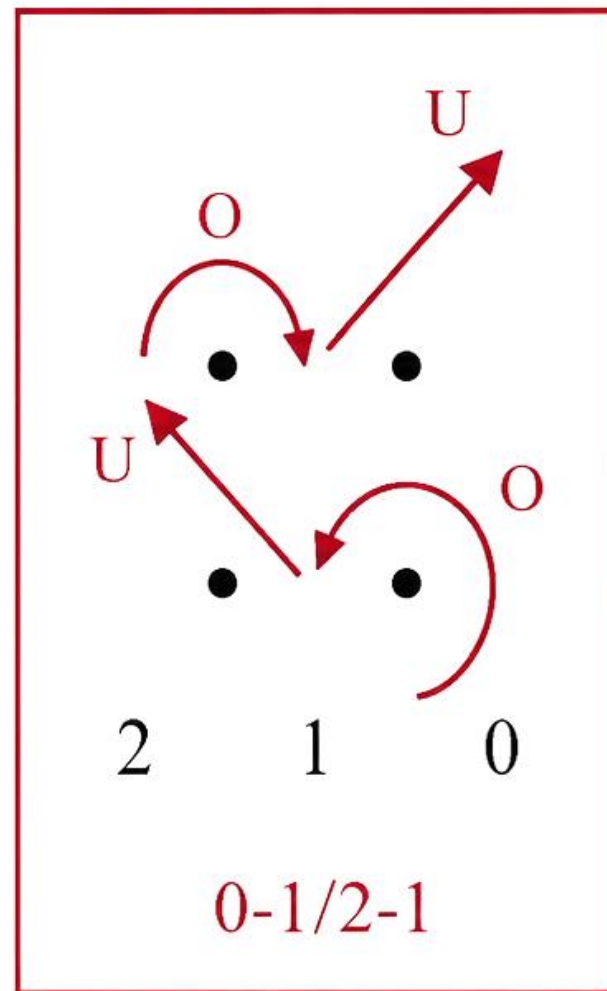
BASIC OVERLAP OR UNDERLAP VARIATIONS

1. Closed Lap: An overlap followed by an underlap in the opposite direction.
2. Open Lap: An overlap followed by an underlap in the same direction.
3. Only Overlap: Only overlap & no underlaps.
4. Only Underlap (Laying-in): Only underlap & no overlaps.
5. Miss Lapping: Neither overlaps nor underlaps.

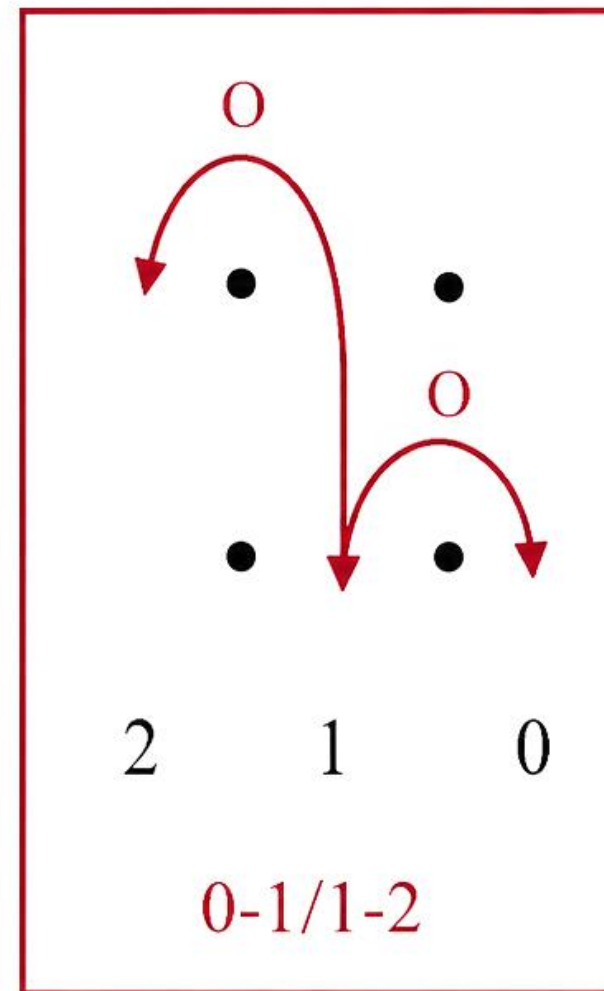
BASIC OVERLAP OR UNDERLAP VARIATIONS



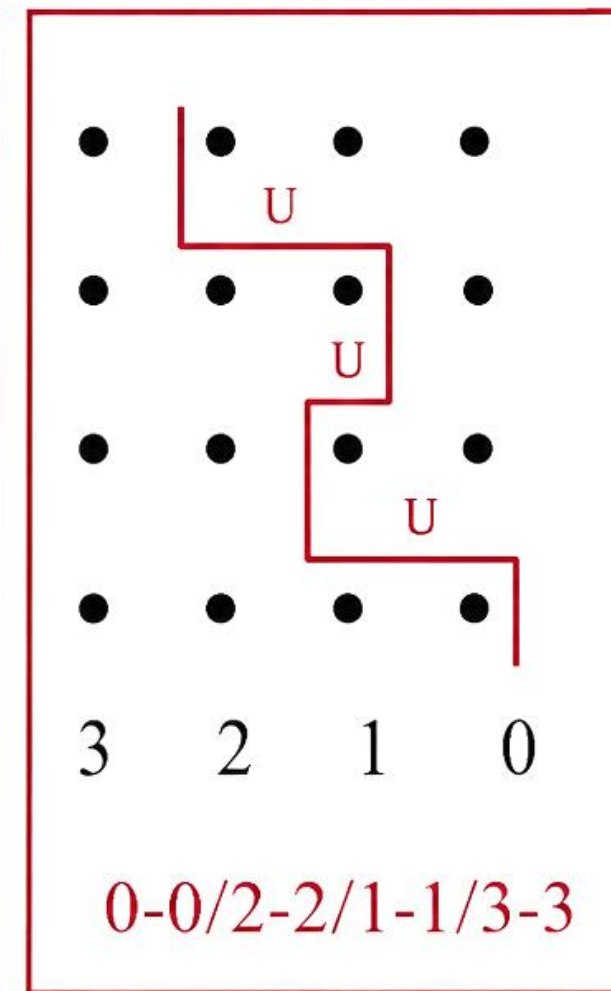
a) Closed lap



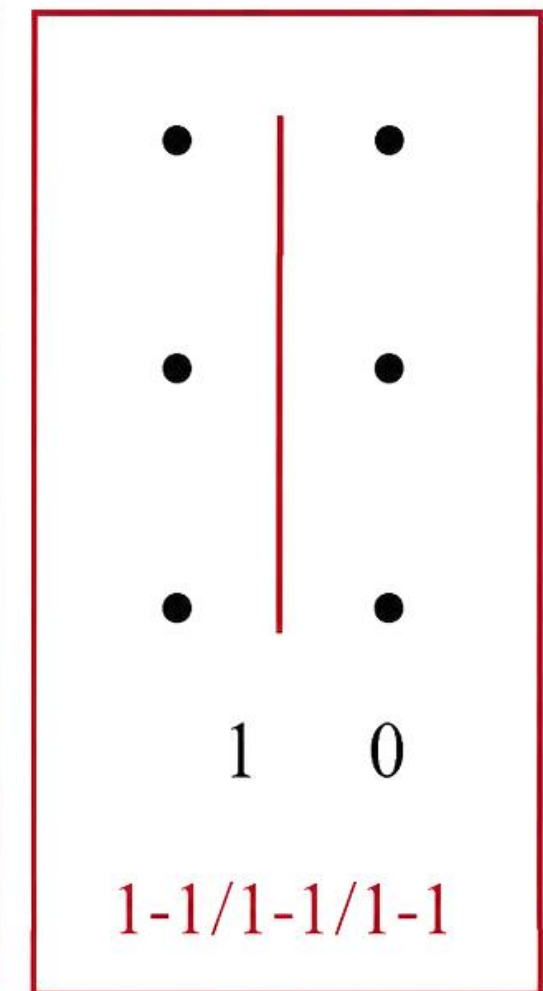
b) Open lap



c) Open lap



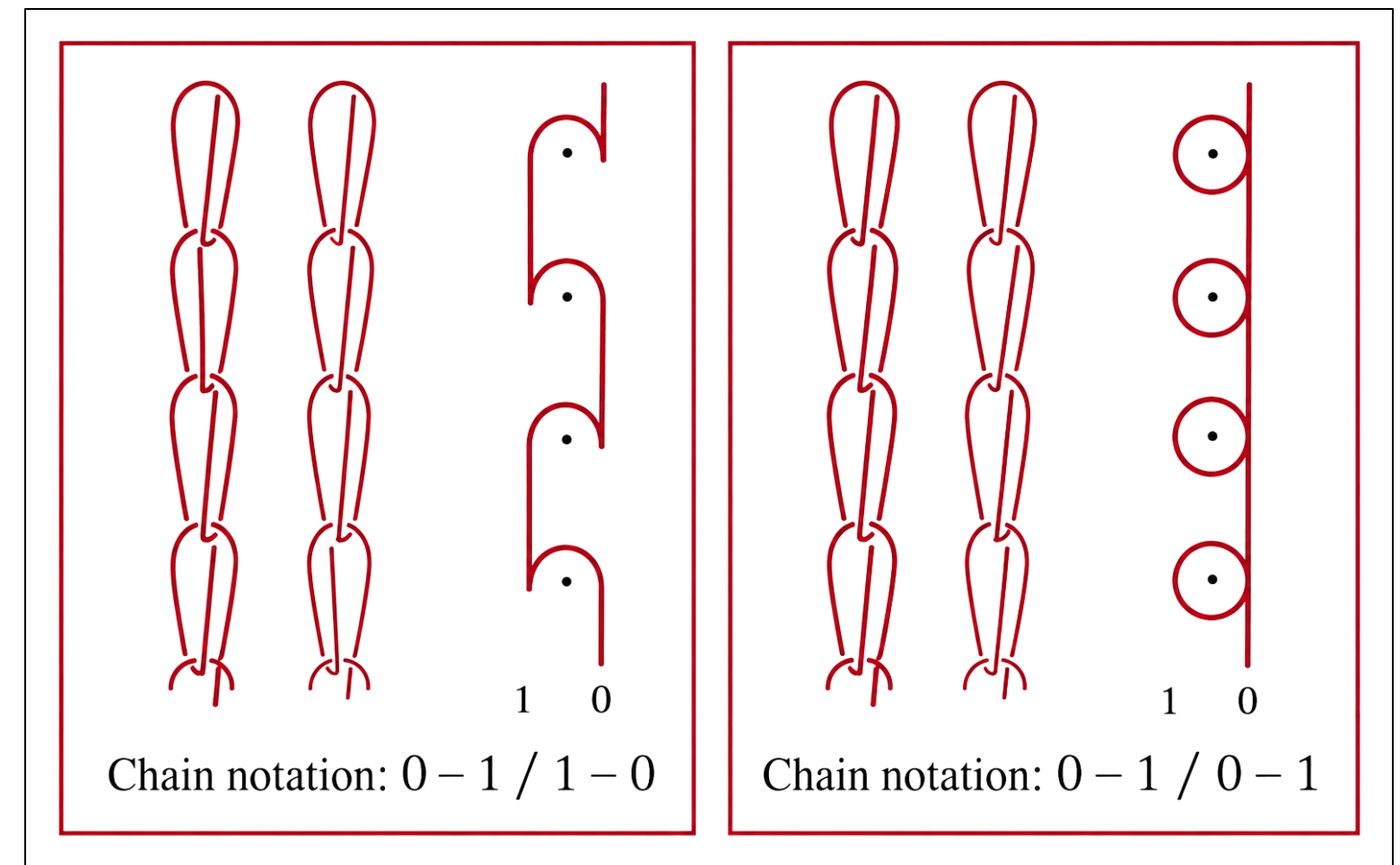
d) Laying-in



e) Miss-lapping

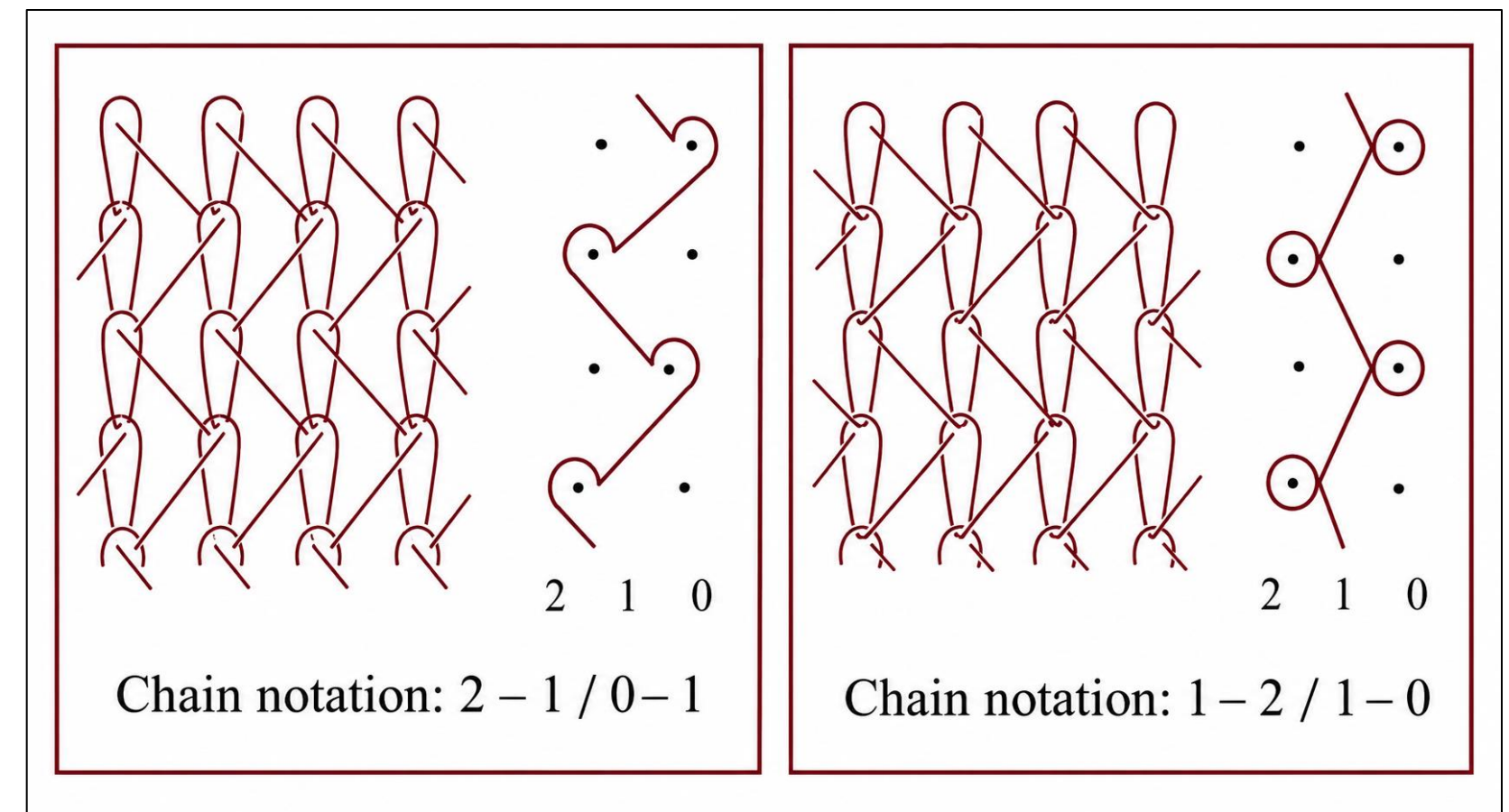
PILLAR OR CHAIN STITCH

- Produced a *vertical chain of consecutively knitted loops on the same needle from the same yarn.*
- The *chain are separated from each other* or connected by other yarns.
- There are *no sideways connection*, so *no fabric* is formed.
- Only wales direction.
- Used to produce *stripe fabric*.



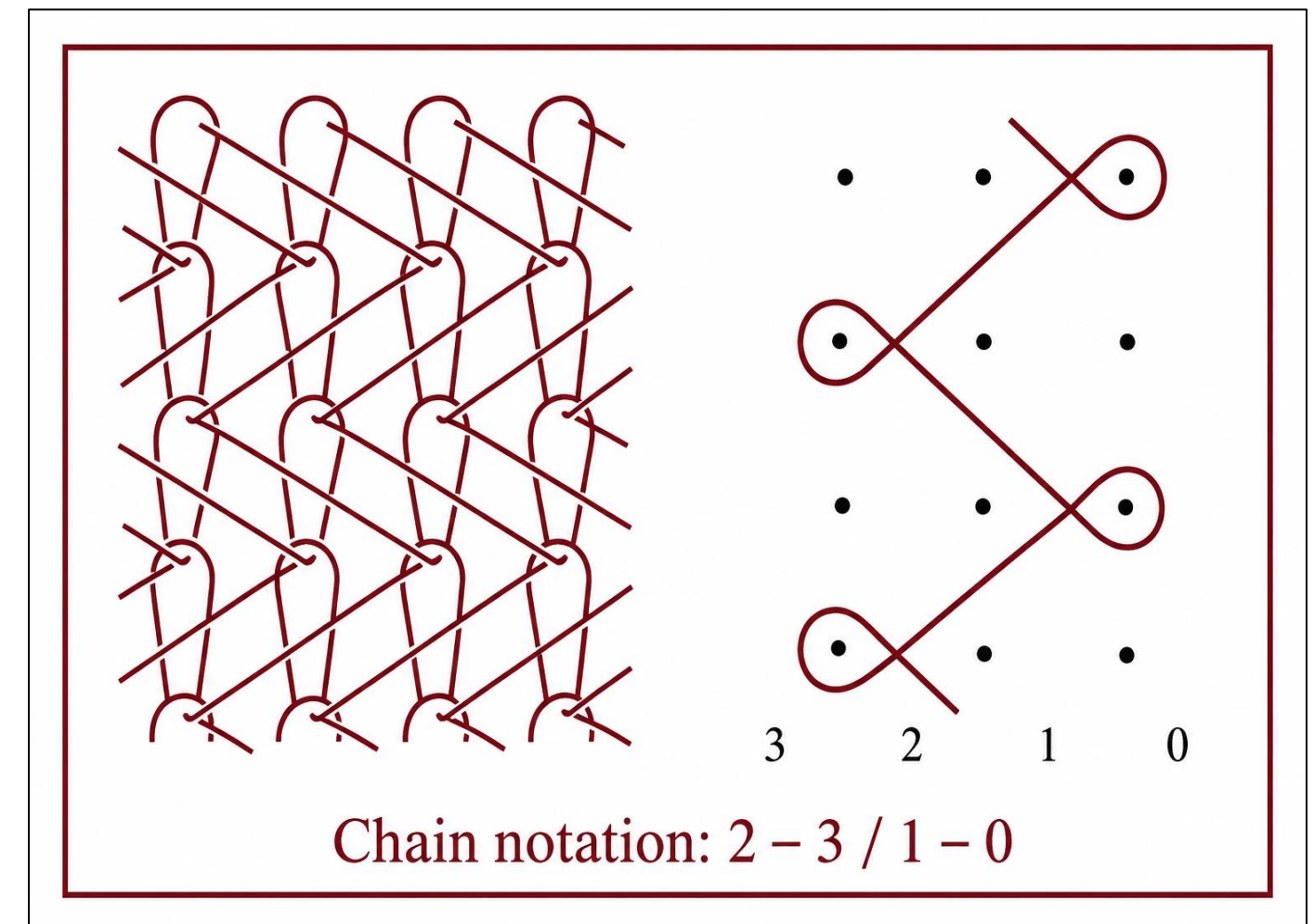
TRICOT STITCH

- Tricot stitch is a stitch formed of one warp; the tricot loops are *disposed in an alternate order in two adjacent wales*.
- The lapping movement is formed by the *guide lapping alternately two adjacent needles*. It is also called a tricot lap and the definition "*1 and 1*" implies an **underlap of one needle space and an overlap of one needle space**.
- *The 1 and 1 movement can be formed open or closed.*
- The closed 1 and 1 structure is more popular.
- The construction is **flexible, light** and very popular in the production of two guide bar fabrics.



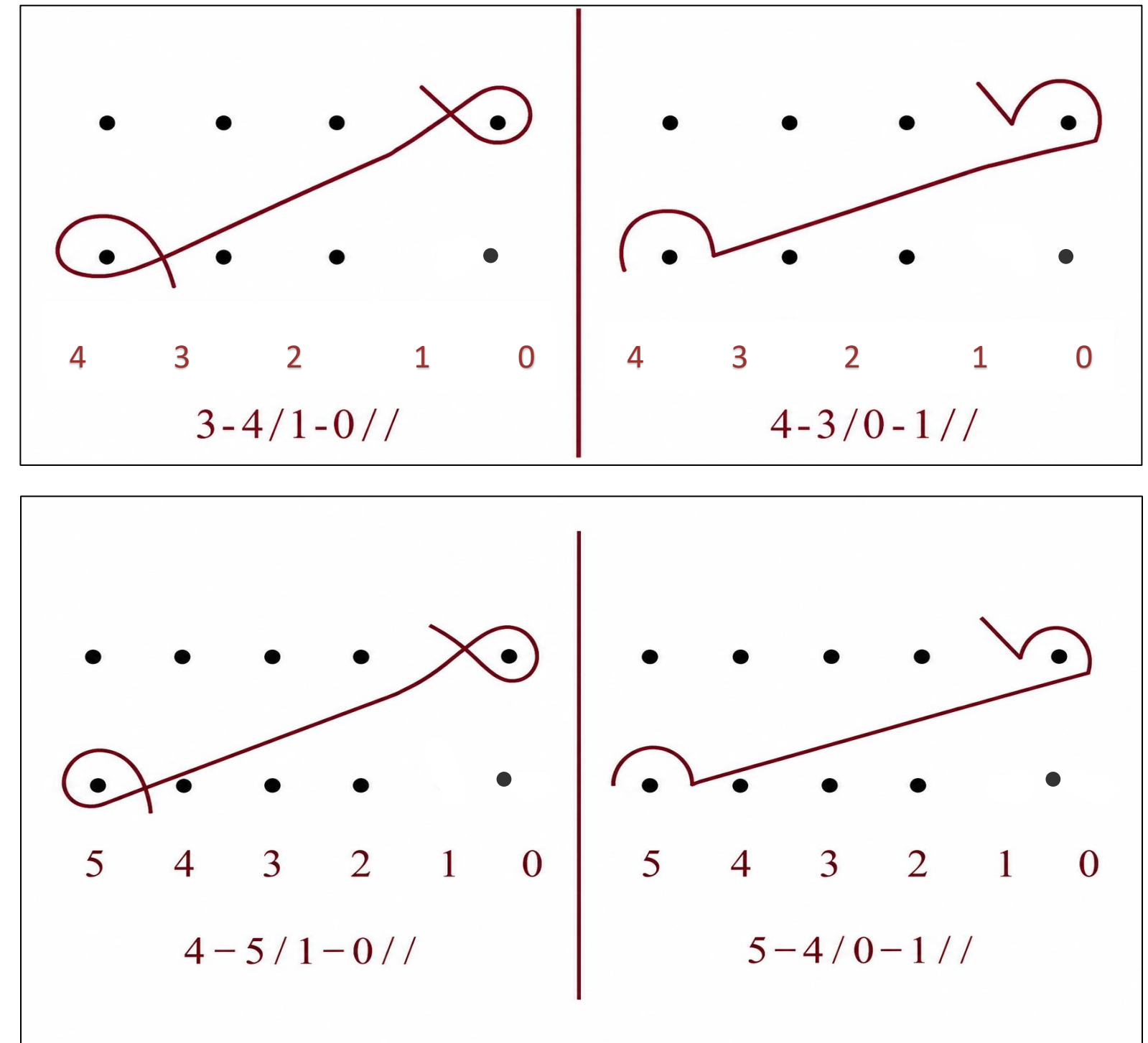
CORD STITCH

- It differs from tricot in the length of links.
- It is knitted from a one warp system and the loops are formed by one yarn in turn in adjacent course, **every two wales**.
- “*2 and 1*” *structure is produced* implies an **underlap** of two needle space and an **overlap** of one needle space.



LONGER RECIPROCATING LAPPING MOVEMENT

- By increasing the underlap shogging movement by one or more needle space (like 3 & 1, 4 & 1) this stitch is formed and it can be produced by open or closed lap.
- These stitches differs from tricot and cord in the length of links.
- *If 3 and 1 lap is constructed then it is called Satin stitch (Above).* More yarn is used with the production of satin stitch. Widthwise stability is increased together with fabric weight and cover factor.
- *If 4 and 1 lap is constructed then it is called Velvet stitch. (Below)*



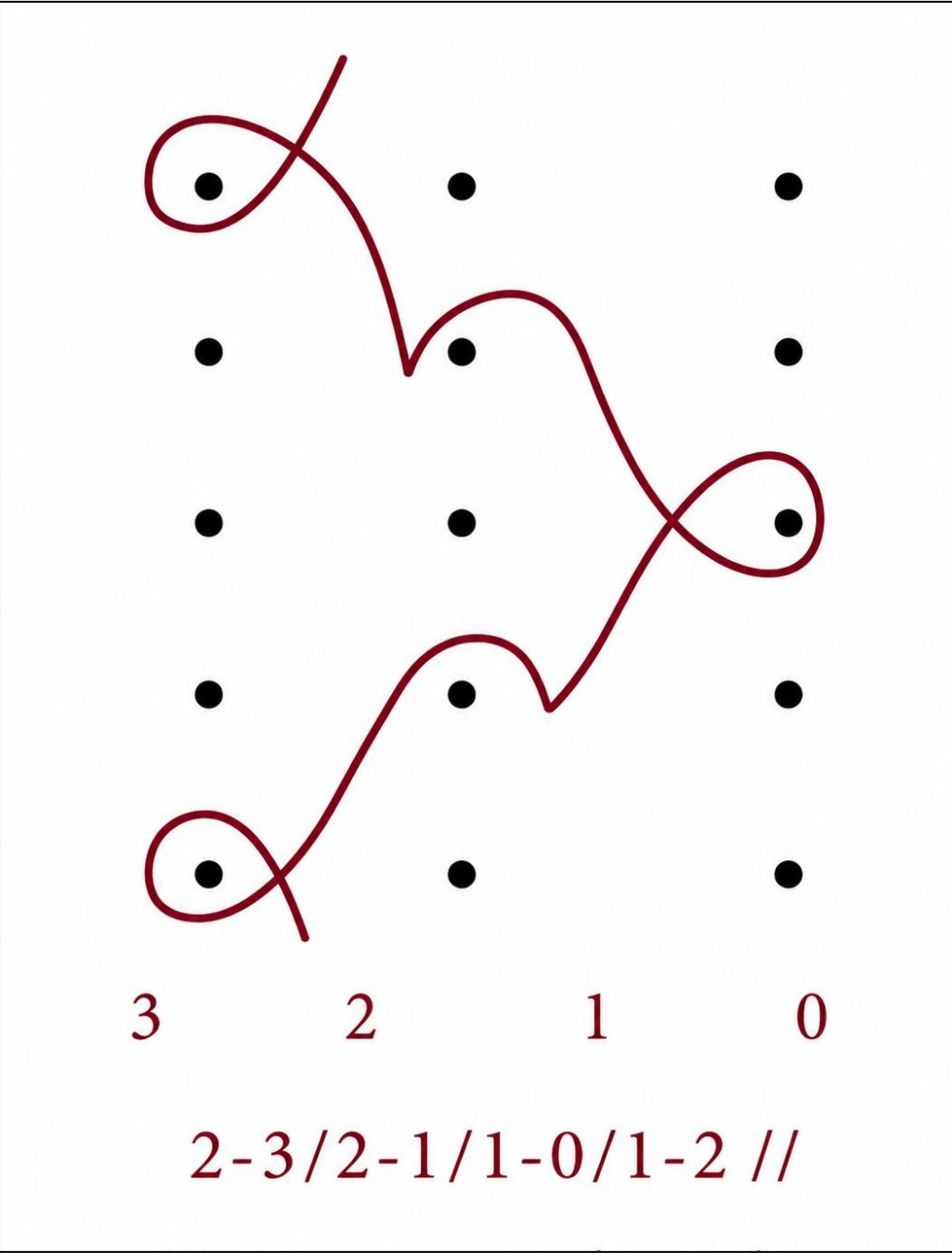
ATLAS STITCH

It is a stitch in which each yarn consecutively forms loops in a **multitude of adjacent wales**. Atlas lapping is often used with the **guide bars threaded with colored yarns**. With two bars moving in opposition, the threading arrangements produce, due to the atlas movement, diagonal or diamond shapes. Atlas stitch is formed by the combination of open and close laps.

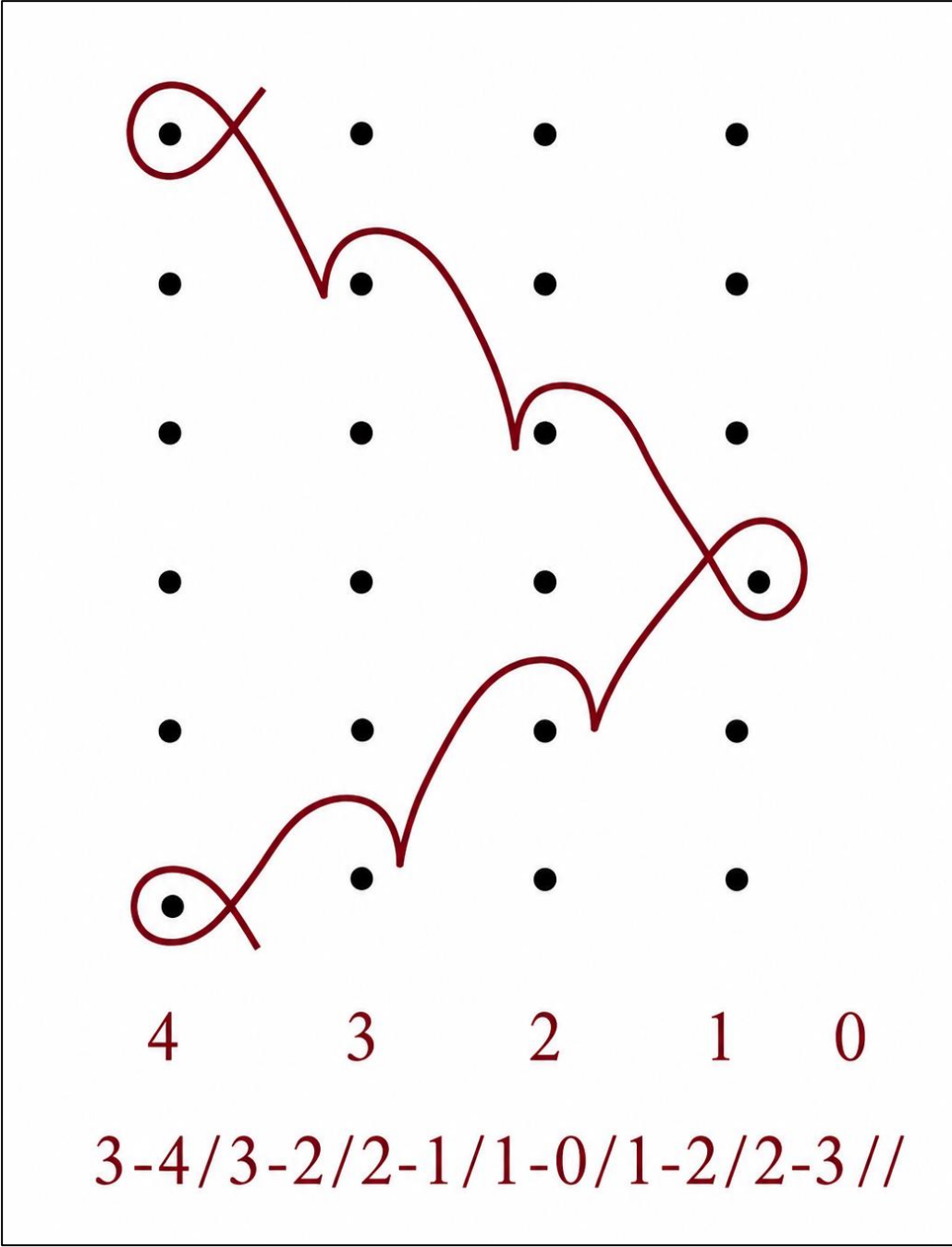
There are various types of atlas:

- (i) Tricot atlas- 2 course tricot atlas, 3 course, 5 course etc.
- (ii) Cord type atlas- It is formed on the **basis of cord stitch**. 5 wales and 4 course in one repeat.
- (iii) Satin type atlas- On the basis of **satin stitch principle**. 7 wales and 4 course in one repeat.
- (iv) Velvet type atlas- On the basis of **velvet stitch principle**. At least 9 wales and 4 course in one repeat.

ATLAS STITCH

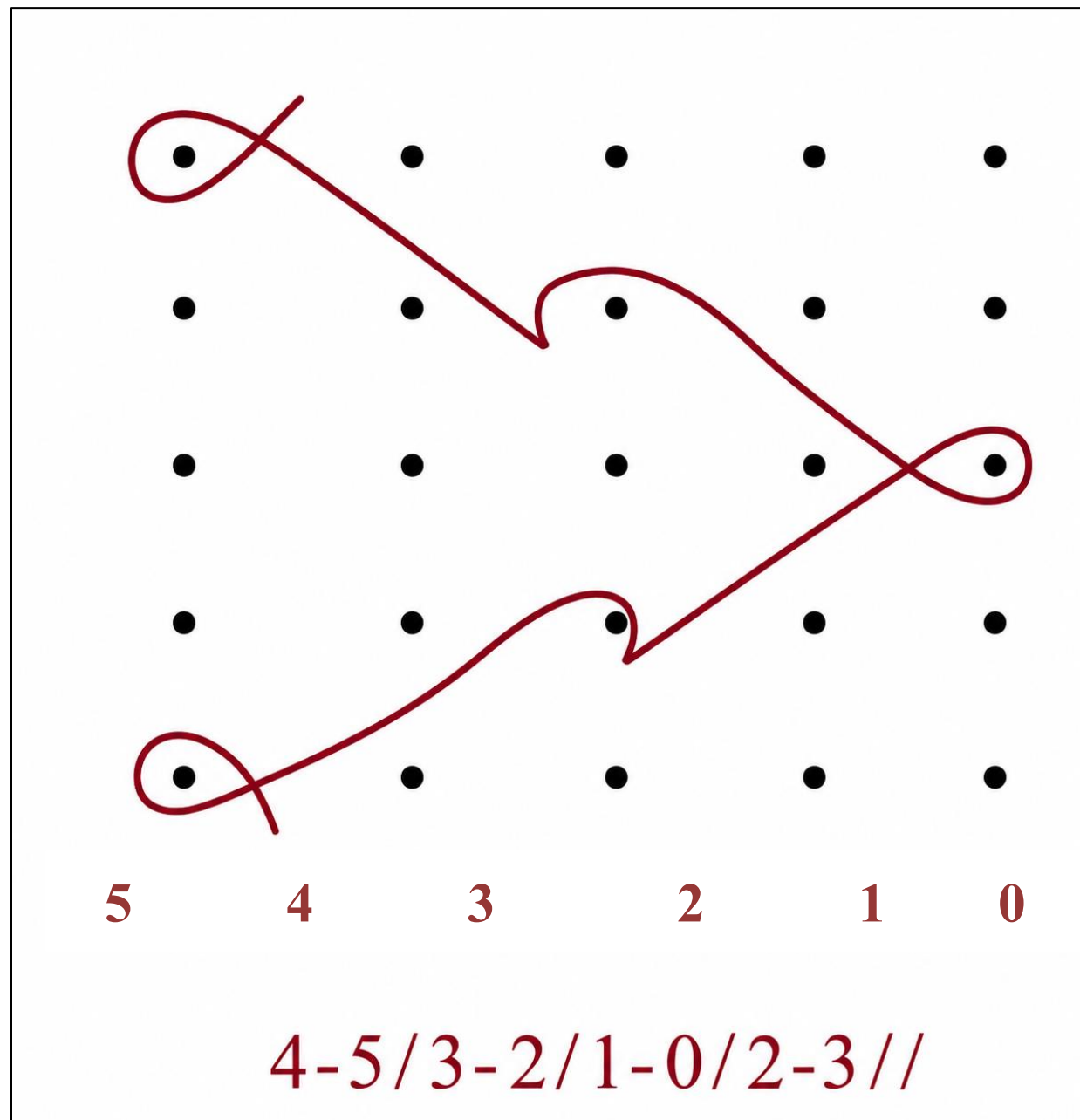


Two Course Tricot Atlas

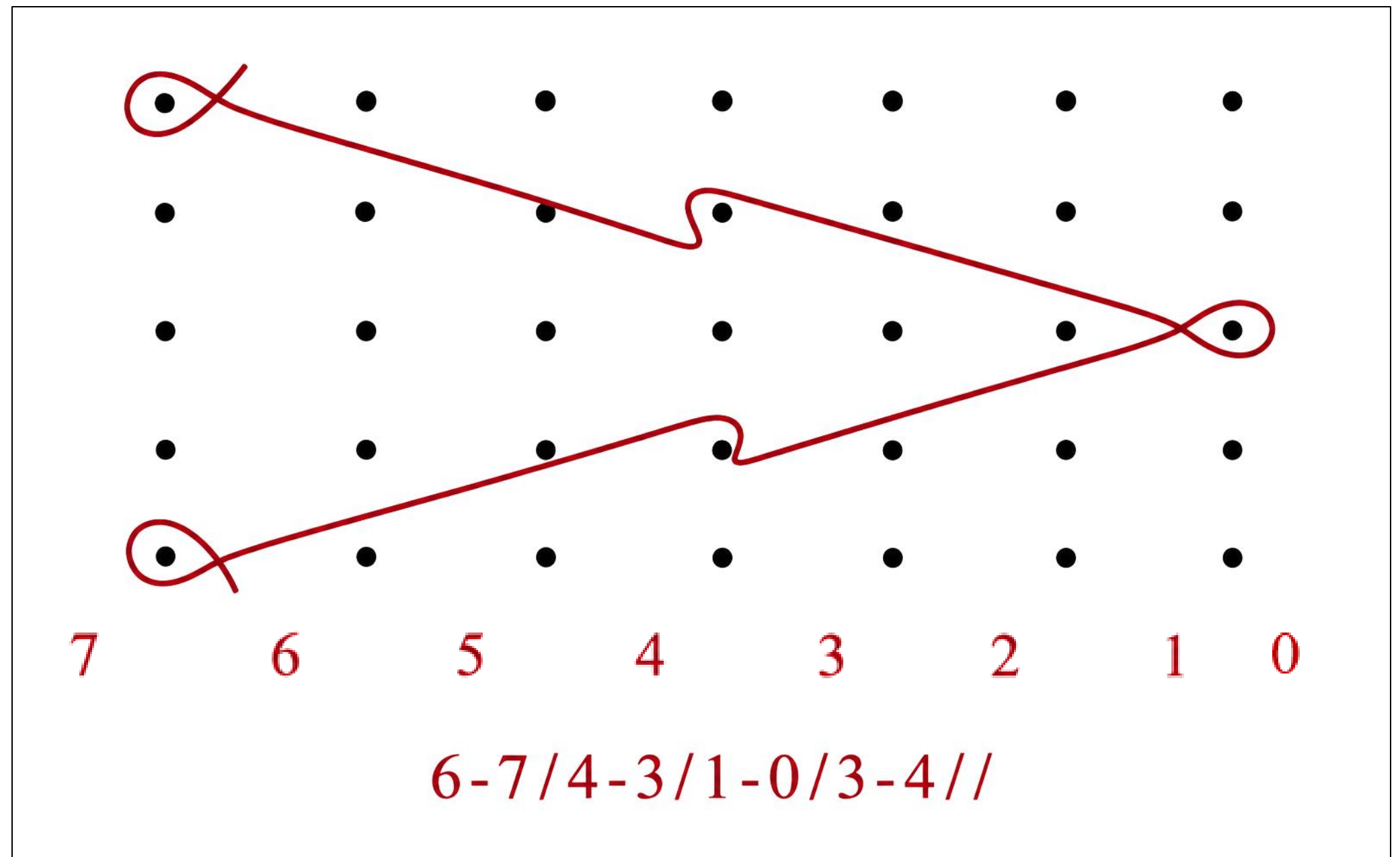


Three Course Tricot Atlas

ATLAS STITCH

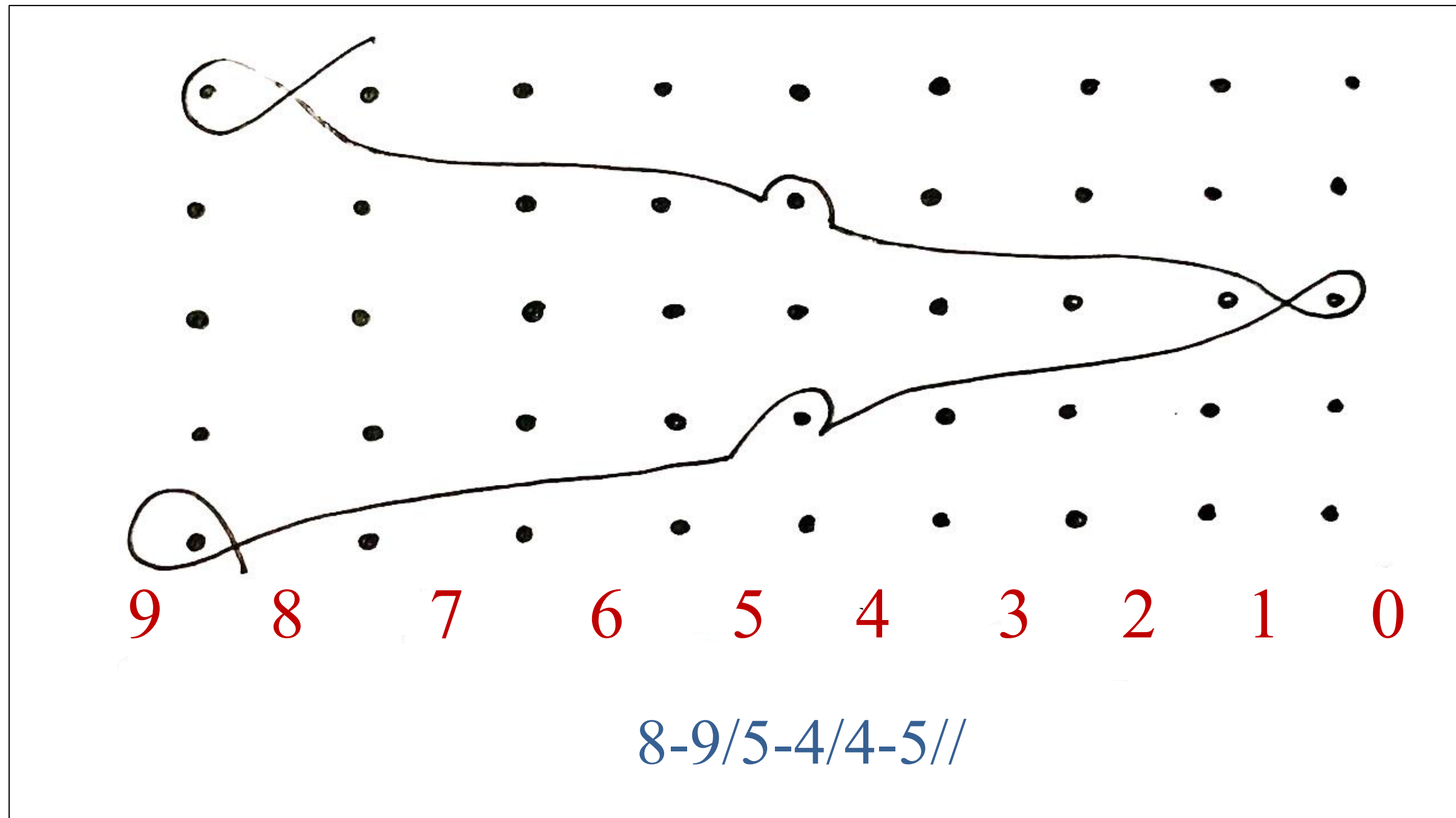


Cord Atlas



Satin Atlas

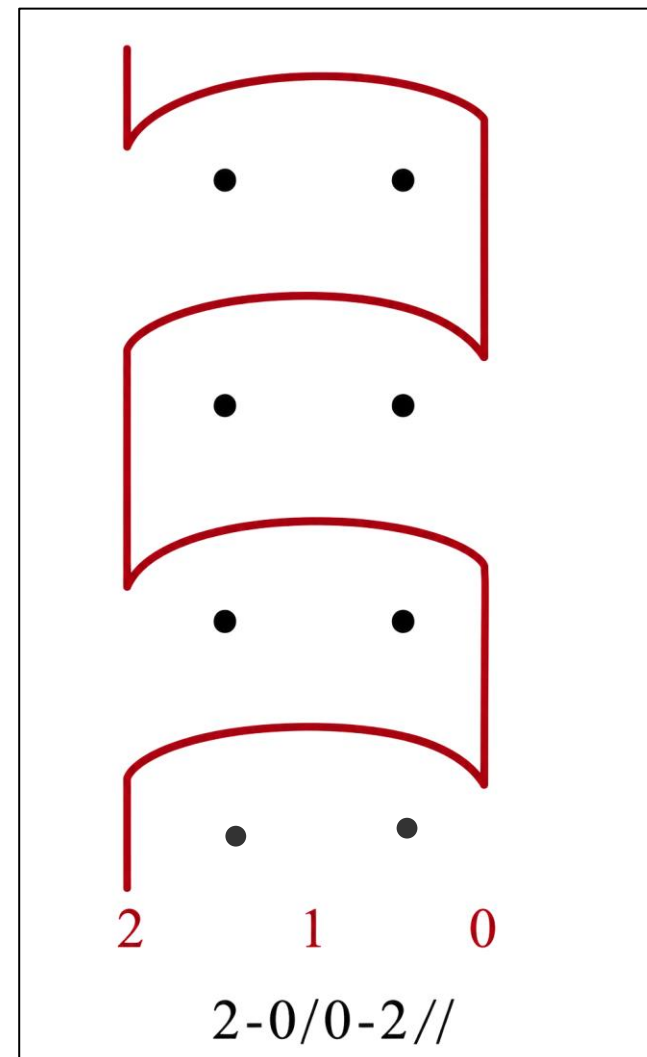
ATLAS STITCH



Velvet Atlas

TWO NEEDLE OVERLAP

- The *guide can be shogged by two needle spaces on the hook side of the needle* during an *overlap*.
- In this way *two needles are wrapped by each yarn* and both will draw the loops simultaneously.
- Used for producing body and stability to a single guide bar fabric.



THANK
YOU !

